GANDHARVA SCHOOL OF MUSIC • OFFICIAL SPECIFICATION

Class Notification & Lifecycle Email Templates Catalog

Complete operational specification, trigger architecture, and visual rendering guide for all automated classroom reminder windows (24h, 12h, 1h, 10m), reschedule events with .ics attachments, cancellation protections, and real-time messaging notifications.

SYSTEM SUBSYSTEM
Email Notification & Reminder Engine

DOCUMENT VERSION
v2.4 Production Release

AUTOMATED SCHEDULERS
Vercel Cron (*/5 * * * *) & Background Workers

TIMEZONE STRATEGY
Dual Zone: Local Recipient Timezone + IST Standard

SECURITY & IDEMPOTENCY
attendanceLogs.reminderLogs deduplication array

GENERATED TIMESTAMP
November 2026 • Authorized Release

Executive Overview

This specification documents the 7 core email templates dispatched across the student and faculty journey. Every template incorporates Gandharva's signature royal purple branding, zero white line borders, high-contrast accessible typography, dynamic timezone localization, and direct one-click classroom entry routes.

1 Notification Engine Architecture & Trigger Matrix

All class reminders are processed via `processUpcomingLessonReminders()` targeting scheduled lessons. Deduplication prevents multiple alerts by logging dispatched windows (`["24h", "12h", "1h", "10m"]`) directly into `Lesson.attendanceLogs.reminderLogs`.

TEMPLATE NAME	TRIGGER EVENT	RECIPIENTS	WINDOW / FILTER	SPECIAL ATTACHMENTS / ACTIONS
1. 24h Reminder	Cron Scheduler (`*/5 * * * *`)	Student & Faculty	Starts between 750m & 1470m ahead	Syllabus review & equipment prep note
2. 12h Reminder	Cron Scheduler (`*/5 * * * *`)	Student & Faculty	Starts between 65m & 750m ahead	Day-of checklist & dual-time badge
3. 1h Reminder	Cron Scheduler (`*/5 * * * *`)	Student & Faculty	Starts between 15m & 65m ahead	Tuning, microphone & audio test checklist
4. 10m Reminder	Cron Scheduler (`*/5 * * * *`)	Student & Faculty	Starts between -5m & +15m ahead	**Green Urgency Badge** • Direct Join Room button
5. Class Rescheduled	Admin/Teacher schedule change	Student & Faculty	Triggered instantly on date/time update	Previous time strikethrough + **updated .ics Calendar invite**
6. Class Cancelled	Admin, Teacher, or Student cancellation	Student & Faculty	Triggered instantly upon status = CANCELLED	Cancellation reason + **Protected Session Balance** assurance
7. New Chat Message	Classroom or Direct Message received	Recipient (when partner absent)	Triggered if partner has not joined live room	Sanitized preview snippet + One-click Live Classroom reply link

> **Timezone Localization Standard:**
> Every template displays the lesson schedule in **two distinct formats**:
> 1. **Recipient Local Time** formatted using `formatInViewerTimezone(startsAt, recipient.timezone)` (e.g., *Wednesday, Nov 11, 2026 at 10:00 AM EST*).
> 2. **Academy Standard Time** formatted using `formatToIST(startsAt)` (e.g., *Nov 11, 2026 at 8:30 PM IST*).

2 Template 1: 24-Hour Class Reminder (Tomorrow)

Trigger Window: 24 hours before class start (`diffMinutes > 750 && diffMinutes <= 1470`) **Idempotency:** `attendanceLogs.reminderLogs.includes("24h")`

Subject (Student): `Reminder: [Discipline] class with [Teacher Name] (Class Tomorrow)`

Subject (Teacher): `Reminder: Teaching session with [Student Name] (Class Tomorrow)`

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Class Tomorrow

24-HOUR REMINDER

Hello **Rohan Mehta**, this is an automated reminder for your upcoming 1-on-1 music class with **Ananya Sharma**.

Discipline / Course: Carnatic Vocals (Beginner Foundation)

Instructor: Ananya Sharma

When: Thursday, November 12, 2026 at 7:00 PM EST

(Indian Standard Time: Nov 13, 2026 at 5:30 AM IST)

Duration: 60 minutes

`Session Code: S0012-SE004`

Your 1-on-1 class is scheduled for tomorrow. Please review any practice notes or sheet music in advance.

Preview Lesson Room →

Quick Checklist:

- Use headphones/earphones for optimal acoustic clarity.
- Tune your instrument before classroom entry.
- Ensure stable Wi-Fi / broadband connection.

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3 Template 2: 12-Hour Class Reminder (Today)

Trigger Window: 12 hours before class start (`diffMinutes > 65 && diffMinutes <= 750`) **Idempotency:** `attendanceLogs.reminderLogs.includes("12h")`

Subject (Student): `Reminder: [Discipline] class with [Teacher Name] (Class Coming Up Today)`

Subject (Teacher): `Reminder: Teaching session with [Student Name] (Class Coming Up Today)`

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Class Coming Up Today

12-HOUR REMINDER

Hello **Rohan Mehta**, this is an automated reminder for your upcoming 1-on-1 music class with **Ananya Sharma**.

Discipline / Course: Carnatic Vocals (Beginner Foundation)

Instructor: Ananya Sharma

When: Today at 7:00 PM EST

(Indian Standard Time: Tomorrow at 5:30 AM IST)

Duration: 60 minutes

Session Code: S0012-SE004

Your lesson is coming up in approximately 12 hours today. Be sure your device and music area are ready.

Check Lesson Room →

Quick Checklist:

- Review previous lesson exercises and teacher feedback.
- Charge your laptop or tablet and position camera at hand/face level.

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4 Template 3 & 4: 1-Hour & 10-Minute Immediate Reminders

• 1-Hour Countdown Window (Amber Urgency)

Trigger Window: 15m to 65m before class start **Subject Line:** `⏰ 1 Hour Reminder: [Discipline] with [Teacher Name]`

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Class Starting in 1 Hour

STARTING IN 1 HOUR

Discipline: Electronic Keyboard • **Duration:** 45 mins

Scheduled Time: Today at 6:00 PM EST (IST: Tomorrow at 4:30 AM IST)

Get Ready & Join →

• 10-Minute Immediate Entry Window (Green Live Alert)

Trigger Window: -5m to +15m of start time **Subject Line:** `🟢 Starting Now: [Discipline] with [Teacher Name]`

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Class Starting in 10 Minutes!

CLASSROOM LIVE NOW

Status: Room is currently active & ready for participants.

Session Link: Direct access active • No PIN required.

Join Classroom Now →

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5 Template 5: Class Rescheduled Notification

Trigger: Admin / Faculty modifies session date or time in Calendar / Enrollment plan

Attachments: RFC 5545 `rescheduled-lesson.ics` calendar event

Subject: `Class Rescheduled: [Discipline] with [Teacher/Student Name]`

Special Action: Resets `attendanceLogs.reminderLogs = []` so 24h, 12h, 1h, 10m reminders fire freshly for the new time

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Class Rescheduled

Hello **Priya Sen**, your upcoming music class with **Vikram Iyer** has been rescheduled by **Academy Administration**.

Discipline / Course: Hindustani Classical Violin

Instructor: Vikram Iyer

Previous Time: ~~Wed, Nov 11, 2026 at 4:00 PM IST~~

New Scheduled Time: Friday, November 13, 2026 at 5:00 PM IST

(Indian Standard Time: Nov 13, 2026 at 5:00 PM IST)

Duration: 60 minutes

Session Code: S0015-SE003

Go to Lesson Room →

An updated calendar invitation (.ics) is attached to this email so your Google Calendar, Outlook, or Apple Calendar stays up to date.

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6 Template 6: Class Cancelled Notification

Trigger:	Admin cancels class, or participant invokes `cancelLessonAction`	**Parties Notified:**	Both Student and Teacher simultaneously
Subject:	`Lesson Cancelled: [Discipline] with [Teacher/Student Name]`		
Business Rule:	Student balance is preserved; cancellation reason is prominently logged		

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Class Cancelled

Hello **Rohan Mehta**, your upcoming music class with **Ananya Sharma** has been cancelled by **Academy Administration**.

Discipline / Course: Carnatic Vocals (Beginner Foundation)

Instructor: Ananya Sharma

Scheduled Time: Thursday, November 12, 2026 at 7:00 PM EST

(IST: Nov 13, 2026 at 5:30 AM IST)

Cancelled By: Academy Administration (admin)

Reason: Instructor attending annual university masterclass examination.

`Session Code: S0012-SE004`

Your class credit remains protected. You can browse other available slots or view your dashboard anytime to schedule your next session.

Go to Student Dashboard →

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7 Template 7: Chat New Message Notification

Trigger:	Direct message sent, or message posted in live classroom when partner hasn't joined	**Security:**	Strict PII filter active (phone/email links restricted)
Subject:	💬 `New Message from [Sender Name] on Gandharva Music`		
Call to Action:	Direct route to `/lesson/[lessonId]` or `/messages?conversationId=[id]`		

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New Message Received

Hello **Rohan Mehta**, you have received a new message from **Ananya Sharma** (teacher):

MESSAGE FROM ANANYA SHARMA:

"I am waiting in the live classroom with the tanpura drone ready. Please join whenever you are ready!"

Open Live Classroom & Reply →

Personal contact details and external links remain restricted in chat for safety.

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End of Specification • Gandharva School of Music Notification Templates • Authorized Release